Task [linux + shell]

<u>Part 1: LVM (Logical Volume Manager):</u> LVM allows flexible disk management by abstracting physical storage into logical volumes that can be resized dynamically.

- Look for the second disk.

```
[root@system ~]# lsblk
NAME    MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda       8:0    0   75G  0 disk
├─sda1    8:1    0    1M  0 part
├─sda2    8:2    0  512M  0 part /boot
├─sda3    8:3    0   25G  0 part /
├─sda4    8:4    0    8G  0 part /home
└─sda5    8:5    0  3.9G  0 part [SWAP]
sdb       8:16   0   20G  0 disk
```

- Partition the second disk for LVM.

```
[root@system ~]# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0x222be9c3.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-41943039, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-41943039, default 4194303
9): +800M

Created a new partition 1 of type 'Linux' and of size 800 MiB.
```

```
Command (m for help): t
Selected partition 1
Hex code or alias (type L to list all): 8e
Changed type of partition 'Linux' to 'Linux LVM'.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

[root@system ~]# lsblk
NAME    MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda       8:0    0   75G  0 disk
├─sda1    8:1    0    1M  0 part
├─sda2    8:2    0  512M  0 part /boot
├─sda3    8:3    0   25G  0 part /
├─sda4    8:4    0    8G  0 part /home
└─sda5    8:5    0  3.9G  0 part [SWAP]
sdb       8:16   0   20G  0 disk
└─sdb1    8:17   0  800M  0 part
```

```
[root@system ~]# pvcreate /dev/sdb1
  Physical volume "/dev/sdb1" successfully created.
  Creating devices file /etc/lvm/devices/system.devices
[root@system ~]# pvs
  PV         VG Fmt  Attr PSize   PFree
  /dev/sdb1     lvm2 ---  800.00m 800.00m
```

- Create a volume group, and set 16M as extends

```
[root@system ~]# vgcreate -s 16M vgdata /dev/sdb1
  Volume group "vgdata" successfully created
[root@system ~]# vgdisplay vgdata
  --- Volume group ---
  VG Name               vgdata
  System ID
  Format                lvm2
  Metadata Areas        1
  Metadata Sequence No  1
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                0
  Open LV               0
  Max PV                0
  Cur PV                1
  Act PV                1
  VG Size               784.00 MiB
  PE Size               16.00 MiB
  Total PE              49
  Alloc PE / Size       0 / 0
  Free  PE / Size       49 / 784.00 MiB
  VG UUID               pQsuMS-xJ8c-9Uaj-akgO-E2Ub-kfK1-GUkzFE
```

.

due to slight rounding or metadata overhead, LVM reports `total PE = 49` and VG size = 784M.

- Divided a volume group containing 50 extends on volume group lv.

```
[root@system ~]# lvcreate -n lv -l 100%VG vgdata
  Logical volume "lv" created.
[root@system ~]# lvs
  LV   VG     Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
  lv   vgdata -wi-a----- 784.00m
```

- Make it as ext4 file system.

mkfs creates a filesystem so the operating system can store and manage files.

```
[root@system ~]# mkfs.ext4 /dev/vgdata/lv
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 200704 4k blocks and 50176 inodes
Filesystem UUID: 5031cfb4-1129-4eb8-aa84-f3ea09e7e1af
Superblock backups stored on blocks:
        32768, 98304, 163840

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done

[root@system ~]# blkid /dev/vgdata/lv
/dev/vgdata/lv: UUID="5031cfb4-1129-4eb8-aa84-f3ea09e7e1af" BLOCK_SIZE="4096" TYPE="ext4"
```

- Mounted automatically under /mnt/data.

Mount attaches a filesystem to the Linux directory tree so it becomes accessible.

`mount` alone → temporary, runtime only

`/etc/fstab` → persistent, survives reboots

```
[root@system ~]# mkdir /mnt/data
[root@system ~]# mount /dev/vgdata/lv /mnt/data
[root@system ~]# mount -a
[root@system ~]# mount | grep /mnt/data
/dev/mapper/vgdata-lv on /mnt/data type ext4 (rw,relatime,seclabel)
```

```
[root@system ~]# vim /etc/fstab
[root@system ~]# mount -a
```

```
#
# /etc/fstab
Tools ed by anaconda on Tue Oct 21 08:25:13 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=8cf60cf6-545c-484f-a118-fea556b8afad /                       xfs     defaults        0 0
UUID=79c151bc-54c1-4359-9eef-a16f934c4479 /boot                   xfs     defaults        0 0
UUID=eacae113-e599-44e8-ace2-8204d432b903 /home                   xfs     defaults        0 0
UUID=8a28f763-17b2-4f21-b337-61c6c833364e none                    swap    defaults        0 0
/dev/vgdata/lv                            /mnt/data               ext4    defaults        0 0
```

To get exactly 50 extents, each 16 MB, I increased the partition size to a little more than 800 MB, because LVM reserves some space for metadata. This ensures the volume group contains the required 50 usable extents.

```
[root@system ~]# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): d
Selected partition 1
Partition 1 has been deleted.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-41943039, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-41943039, default 4194303
9): +2G

Created a new partition 1 of type 'Linux' and of size 2 GiB.

Command (m for help): w
```

```
[root@system ~]# vgcreate -s 16M vgdata /dev/sdb1
  Physical volume "/dev/sdb1" successfully created.
  Volume group "vgdata" successfully created
[root@system ~]# lvcreate -n lv -l 50 vgdata
WARNING: ext4 signature detected on /dev/vgdata/lv at offset 1080. Wipe it? [
y/n]: y
  Wiping ext4 signature on /dev/vgdata/lv.
  Logical volume "lv" created.
```

After fixing the partition size, I repeated the same steps used earlier: I initialized the physical volume, recreated the volume group with a 16 MB extent size, created the logical volume with exactly 50 extents, formatted it with ext4, mounted it on /mnt/data, and finally verified the changes using `lsblk`, `vgdisplay`, `lvdisplay`, and mount checks.

```
[root@system ~]# vgdisplay vgdata
  --- Volume group ---
  VG Name               vgdata
  System ID
  Format                lvm2
  Metadata Areas        1
  Metadata Sequence No  2
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                1
  Open LV               1
  Max PV                0
  Cur PV                1
  Act PV                1
  VG Size               1.98 GiB
  PE Size               16.00 MiB
  Total PE              127
  Alloc PE / Size       50 / 800.00 MiB
  Free  PE / Size       77 / 1.20 GiB
  VG UUID               bO1cnx-GeYk-1eQX-RXT2-G7ul-uYEI-df2fMs
```

Part 2: users, groups and permissions

1.

- Add user: user1.
- set uid=601.
- Password: redhat.
- The user's login shell should be non-interactive. (no ssh access to server)

```
[root@system ~]# useradd -u 601 -s /sbin/nologin user1
useradd warning: user1's uid 601 outside of the UID_MIN 5000 and UID_MAX 6000
0 range.
[root@system ~]# id user1
uid=601(user1) gid=6004(user1) groups=6004(user1)
```

```
[root@system ~]# useradd -u 601 -s /sbin/nologin user1 2>/dev/null
[root@system ~]# passwd user1
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
```

```
[root@system ~]# su - user1
This account is currently not available.
```

2.

- Add user1 to group TrainingGroup.

```
[root@system ~]# groupadd TrainingGroup
[root@system ~]# usermod -aG TrainingGroup user1
[root@system ~]# id user1
uid=601(user1) gid=6004(user1) groups=6004(user1),6005(TrainingGroup)
```

3.

- Add users: user2, user3.
- The Additional group of the two users: user2, user3 is the admin group Password: redhat.
- user 3 with root permissions.

```
[root@system ~]# groupadd admin
```

```
[root@system ~]# useradd -G admin user2
[root@system ~]# useradd -G admin user3
[root@system ~]# passwd user2
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
[root@system ~]# passwd user3
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
[root@system ~]# usermod -aG wheel user3
```

`wheel` is a special system group that allows users to run commands as root using `sudo`.

```
[root@system ~]# su - user3
Last login: Sun Nov 23 16:46:20 EET 2025 on pts/0
[user3@system ~]$ sudo whoami
[sudo] password for user3:
root
```

```
[root@system ~]# id user3
uid=6005(user3) gid=6008(user3) groups=6008(user3),10(wheel),6006(admin)
```

<u>Part 3: SSH</u> (Secure Shell) It is a secure way to connect to another Linux server remotely.

- Generate SSH key.
- connect to different VM without password.

```
[root@system ~]# su - user3
Last login: Sun Nov 23 19:29:39 EET 2025 on pts/1
[user3@system ~]$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/user3/.ssh/id_rsa):
/home/user3/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase for "/home/user3/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/user3/.ssh/id_rsa
Your public key has been saved in /home/user3/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:2n+QMrx1a+CcVtAPZavnhrz3tBTgkMqEUdByaw/bVxQ user3@system.example.com
The key's randomart image is:
+---[RSA 4096]----+
|        o+.     E.|
|        .oo  . o. |
|        .o..+ +.. |
|         o+o = o. |
|        .So=o =.. |
|        o+.=o+.o .|
|       . .B B.= ..|
|         ..= * =..|
|          ..o.+ o.|
+----[SHA256]-----+
```

```
[user3@system ~]$ ssh-copy-id ldap@ipaserver.example.com
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/user3/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
(ldap@ipaserver.example.com) Password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'ldap@ipaserver.example.com'"
and check to make sure that only the key(s) you wanted were added.

[user3@system ~]$ ssh ldap@ipaserver.example.com
Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Thu Nov 27 09:17:24 2025 from 192.168.99.10
[ldap@ipaserver ~]$ exit
```

For this task, we used `user3` instead of root because logging in as root over SSH is insecure. `user3` is a normal user with sudo privileges, which allows performing necessary administrative tasks safely. We generated an SSH key with `ssh-keygen -t rsa -b 4096` (`-t rsa` specifies the RSA key type, `-b 4096` sets the key length to 4096 bits for strong security). The public key was copied to the remote IPA server (`ldap@ipaserver.example.com`) using `ssh-copy-id`, enabling passwordless SSH login.

Part 4: permissions

- Copy /etc/fstab to /var/tmp name admin. (File System Table) defines how and where filesystems are mounted automatically at boot

```
[root@system ~]# cp /etc/fstab /var/tmp/admin
[root@system ~]# chown user1:user1 /var/tmp/admin
[root@system ~]# chmod 700 /var/tmp/admin
[root@system ~]# ls -l /var/tmp/admin
-rwx------. 1 user1 user1 806 Nov 23 20:35 /var/tmp/admin
```

- user1 could read, write and modify it.

```
[user1@system ~]$ cd /var/tmp
[user1@system tmp]$ ls
admin
```

```
[user1@system tmp]$ ls -l admin
-rwx------. 1 user1 user1 806 Nov 23 20:35 admin
[user1@system tmp]$ echo "test" >> admin
[user1@system tmp]$ cat admin

#
# /etc/fstab
# Created by anaconda on Tue Oct 21 08:25:13 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=8cf60cf6-545c-484f-a118-fea556b8afad /                       xfs     def
aults        0 0
UUID=79c151bc-54c1-4359-9eef-a16f934c4479 /boot                   xfs     def
aults        0 0
UUID=eacae113-e599-44e8-ace2-8204d432b903 /home                   xfs     def
aults        0 0
UUID=8a28f763-17b2-4f21-b337-61c6c833364e none                    swap    def
aults        0 0
/dev/vgdata/lv                            /mnt/data               ext4    def
aults        0 0
test
```

- user2 can't do any permission.

```
[user2@system ~]$ cd /var/tmp
[user2@system tmp]$ cat admin
cat: admin: Permission denied
[user2@system tmp]$ echo "test" >> admin
-bash: admin: Permission denied
```

Part 5: permissions

Security-Enhanced Linux: SELinux is a mandatory access control system that adds an extra layer of security to Linux. Enforcing blocks access, Permissive logs only, Disabled turns SELinux off.

- SELinux must be running in the Enforcing mode (permanent even after reboot).

```
[root@system ~]# vim /etc/selinux/config
```

```
# This file controls the state of SELinux on the system.
# SELINUX= can take one of these three values:
#     enforcing - SELinux security policy is enforced.
#     permissive - SELinux prints warnings instead of enforcing.
#     disabled - No SELinux policy is loaded.
# See also:
# https://docs.fedoraproject.org/en-US/quick-docs/getting-started-with-selinux/#getting-star
#
# NOTE: In earlier Fedora kernel builds, SELINUX=disabled would also
# fully disable SELinux during boot. If you need a system with SELinux
# fully disabled instead of SELinux running with no policy loaded, you
# need to pass selinux=0 to the kernel command line. You can use grubby
# to persistently set the bootloader to boot with selinux=0:
#
#    grubby --update-kernel ALL --args selinux=0
#
# To revert back to SELinux enabled:
#
#    grubby --update-kernel ALL --remove-args selinux
#
SELINUX=enforcing
# SELINUXTYPE= can take one of these three values:
#     targeted - Targeted processes are protected,
#     minimum - Modification of targeted policy. Only selected processes are protected.
#     mls - Multi Level Security protection.
SELINUXTYPE=targeted


~
~
~
~
~
~
:wq
```

Then systemctl reboot

```
[root@system ~]# getenforce
Enforcing
[root@system ~]# sestatus
SELinux status:                 enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:             targeted
Current mode:                   enforcing
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Memory protection checking:     actual (secure)
Max kernel policy version:      33
[root@system ~]#
```

If use setenforce 1 # 1 = enforcing, 0 = permissive → This changes SELinux mode immediately, but won't persist after reboot.

Part 6: bash script and processes

- Write a shell script that will keep running for 10 mins in the background.
- check the process that it's created and try to kill using commands.

```
[root@system ~]# vim monitor_process.sh
[root@system ~]# chmod +x monitor_process.sh
[root@system ~]# ./monitor_process.sh
Started background process with PID: 235646
Process 235646 is still running. Attempting to kill it...
Process 235646 killed.
./monitor_process.sh: line 21: 235646 Killed                  sleep 600
Process 235646 has already exited.
Process 235646 has already exited.
10 minutes are over. Script finished.
```

ps → check if process exists

```
#!/bin/bash
# monitor_process.sh

# Start a long-running process in the background (example: sleep 600)
sleep 600 &
PID=$!
echo "Started background process with PID: $PID"

# Run the script for 10 minutes
START_TIME=$(date +%s)
END_TIME=$((START_TIME + 600))

while [ $(date +%s) -lt $END_TIME ]; do
    if ps -p $PID > /dev/null; then
        echo "Process $PID is still running. Attempting to kill it..."
        kill -9 $PID && echo "Process $PID killed."
    else
        echo "Process $PID has already exited."
    fi
    sleep 10  # prevent CPU spike
done

echo "10 minutes are over. Script finished."
```

Part 7: Yum Repo

- Install tmux on your machine.

`tmux` is a terminal multiplexer: run multiple terminal sessions in one window. Can detach and reattach sessions. Useful for long-running tasks like installing packages or building a repo

```
[root@system ~]# yum install -y tmux
Last metadata expiration check: 0:42:44 ago on Thu 27 Nov 2025 12:49:34 PM EE
T.
Package tmux-3.3a-13.20230918gitb202a2f.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

- Install apache server on your machine(httpd) and Install mysql.
  httpd (Apache) → will host local repo (Zabbix RPMs) via HTTP
  MariaDB → database server (needed for Zabbix later)

```
[root@system ~]# yum install -y httpd
Last metadata expiration check: 0:26:26 ago on Thu 27 Nov 2025 09:53:24 PM EE
T.
Package httpd-2.4.63-1.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@system ~]# systemctl enable --now httpd
[root@system ~]# yum install -y mariadb mariadb-server
Last metadata expiration check: 0:26:40 ago on Thu 27 Nov 2025 09:53:24 PM EE
T.
Package mariadb-3:10.11.11-1.el10.x86_64 is already installed.
Package mariadb-server-3:10.11.11-1.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@system ~]# systemctl enable --now mariadb
```

- Create a local yum repository on your local machine(available publicly) with the zabbix rpms: https://repo.zabbix.com/zabbix/4.4/rhel/7/x86_64/.

```
[root@system ~]# mkdir -p /var/www/html/zabbix
[root@system ~]# cd /var/www/html/zabbix
```

  curl → command-line tool to **download or transfer data** from a URL
  `-L` → **follow redirects** if the URL points to another location
  `-O` → **save the file with the same name** as in the URL

```
[root@system zabbix]# curl -LO https://repo.zabbix.com/zabbix/4.4/rhel/7/x86_
64/zabbix-server-mysql-4.4.0-1.el7.x86_64.rpm
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Curre
nt
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--
  1 2490k    1 32768    0     0  23467      0  0:01:48  0:00:01  0:01:47 2347
 48 2490k   48 1199k    0     0   452k      0  0:00:05  0:00:02  0:00:03  452
 75 2490k   75 1871k    0     0   550k      0  0:00:04  0:00:03  0:00:01  550
100 2490k  100 2490k    0     0   653k      0  0:00:03  0:00:03 --:--:--  653
k
[root@system zabbix]# curl -LO https://repo.zabbix.com/zabbix/4.4/rhel/7/x86_
64/zabbix-agent-4.4.0-1.el7.x86_64.rpm
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Curre
nt
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--
100  418k  100  418k    0     0   273k      0  0:00:01  0:00:01 --:--:--  273
k
[root@system zabbix]# curl -LO https://repo.zabbix.com/zabbix/4.4/rhel/7/x86_
64/zabbix-web-4.4.0-1.el7.noarch.rpm
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Curre
nt
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--
 40 2977k   40 1199k    0     0   640k      0  0:00:04  0:00:01  0:00:03  640
100 2977k  100 2977k    0     0  1059k      0  0:00:02  0:00:02 --:--:-- 1059
k
[root@system zabbix]# curl -LO https://repo.zabbix.com/zabbix/4.4/rhel/7/x86_
64/zabbix-web-mysql-4.4.0-1.el7.noarch.rpm
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Curre
nt
```

```
[root@system zabbix]# createrepo /var/www/html/zabbix
Directory walk started
Directory walk done - 4 packages
Temporary output repo path: /var/www/html/zabbix/.repodata/
Pool started (with 5 workers)
Pool finished
```

```
[root@system zabbix]# firewall-cmd --add-service=http --permanent
Warning: ALREADY_ENABLED: http
success
[root@system zabbix]# firewall-cmd --reload
success
[root@system zabbix]# systemctl restart httpd
```

```
[root@system zabbix]# curl -I http://localhost/zabbix/
HTTP/1.1 200 OK
Date: Thu, 27 Nov 2025 20:44:12 GMT
Server: Apache/2.4.63 (Red Hat Enterprise Linux)
Content-Type: text/html;charset=ISO-8859-1
```

```
[root@system zabbix]# vim /etc/yum.repos.d/local-zabbix.repo
[root@system zabbix]# cat /etc/yum.repos.d/local-zabbix.repo
[local-zabbix]
name=Local Zabbix Repository
baseurl=http://localhost/zabbix
enabled=1
gpgcheck=0

[root@system zabbix]# yum repolist
repo id                                   repo name
AppStream                                 AppStream-Repo
BaseOs                                    BaseOS-Repo
local-zabbix                              Local Zabbix Repository
```

- Disable all other repositories and keep only the new repo.
  Every repository has a `.repo` file in `/etc/dnf/repos.d/`.I can disable them by editing the files or using commands:

```
[root@system zabbix]# dnf config-manager --disable AppStream
```

```
[root@system zabbix]# dnf config-manager --disable BaseOs
```

```
[root@system zabbix]# dnf repolist enabled
repo id                                   repo name
local-zabbix                              Local Zabbix Repository
[root@system zabbix]#
```

- Install zabbix rpms from the new repo (Download zabbix, zabbix-web,php, zabbix-server,zabbix-agent rpm's and their dependencies).

```
[root@system zabbix]# dnf install zabbix-server-mysql zabbix-web-mysql zabbix
-agent
Last metadata expiration check: 0:00:47 ago on Thu 27 Nov 2025 11:44:38 PM EE
T.
```

```
Last metadata expiration check: 0:02:27 ago on Fri 05 Dec 2025 06:31:46 AM EE
T.
Package zabbix-agent-7.0.21-release2.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

Part 8: Network management

- Open port 443 , 80.

  Port 80: HTTP (HyperText Transfer Protocol). Used for normal, unencrypted web traffic. Port 443: HTTPS (HTTP Secure). Used for secure, encrypted web traffic. Opening them allows users to access websites hosted on the server.

  A firewall is a security system that controls and filters network traffic to protect servers and networks.

- Make the changes permanent.

```
[root@system ~]# firewall-cmd --permanent --add-service=http
success
```

```
[root@system ~]# firewall-cmd --permanent --add-service=https
success
[root@system ~]# firewall-cmd --reload
success
```

```
[root@system ~]# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3 enp0s8
  sources:
  services: cockpit dhcpv6-client http https ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

- Block ssh connection for your colleague ip to the VM.
  To prevent unauthorized or restricted users from accessing the server via SSH

```
[root@system ~]# firewall-cmd --permanent --add-rich-rule='rule family="ipv4"
 source address="192.168.1.100" port protocol="tcp" port="22" reject'
success
[root@system ~]# firewall-cmd --reload
success
[root@system ~]# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3 enp0s8
  sources:
  services: cockpit dhcpv6-client http https ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
        rule family="ipv4" source address="192.168.1.100" port port="22" prot
ocol="tcp" reject
```

Part 9: Cronjob

- Create a cronjob that will run at 1:30 AM every day.
  Cron is a Linux scheduler that automatically runs commands or scripts at specified times or intervals
- collect the users logged in and save them in a file. Format : timestamp – users.

Note: the cronjob can run a script.

```
[root@system ~]# vim /usr/local/bin/log_users.sh
[root@system ~]# chmod 0755 /usr/local/bin/log_users.sh
[root@system ~]# vim /var/log/logged_users.log
[root@system ~]# chmod 0644 /var/log/logged_users.log
[root@system ~]# /usr/local/bin/log_users.sh
```

```
[root@system ~]# crontab -e
no crontab for root - using an empty one
crontab: installing new crontab
[root@system ~]# crontab -l
30 1 * * * /usr/local/bin/log_users.sh
```

`tr` cleans up spaces, `cut` extracts the first column (username), and `sed` removes trailing spaces, producing a clean list of unique logged-in users.

```
#!/bin/bash
# /usr/local/bin/log_users.sh
LOGFILE="/var/log/logged_users.log"
DATE=/usr/bin/date
WHO=/usr/bin/who
TR=/usr/bin/tr
CUT=/usr/bin/cut
SORT=/usr/bin/sort
UNIQ=/usr/bin/uniq
SED=/usr/bin/sed

TIMESTAMP=$($DATE "+%Y-%m-%d %H:%M:%S")
# get first column from who (collapse spaces, cut field1), sort & uniq, join on one line
USERS=$($WHO | $TR -s ' ' | $CUT -d' ' -f1 | $SORT | $UNIQ | $TR '\n' ' ')
```

```
echo "$TIMESTAMP - $USERS" >> "$LOGFILE"
```

```
[root@system ~]# cat /var/log/logged_users.log
2025-11-27 16:56:53 - root yasmin
2025-11-27 16:58:02 - root yasmin
2025-11-28 01:30:02 - yasmin
```

Part 10: Mariadb

- install mariadb from the local repo that was created earlier.

```
[root@system zabbix]# dnf config-manager --enable local-zabbix
[root@system zabbix]# dnf install mariadb-server mariadb -y
Local Zabbix Repository                         344 kB/s | 1.5 kB     00:00
Package mariadb-server-3:10.11.11-1.el10.x86_64 is already installed.
Package mariadb-3:10.11.11-1.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

- open ports in iptables from mariadb.
  Allow inbound connections to port 3306/tcp in the firewall on the server where MariaDB is running.
  This can be done using either `firewall` or `iptables`.
  To allow remote or local clients to connect to MariaDB/MySQL database.

```
[root@system zabbix]# systemctl enable --now mariadb
[root@system zabbix]# firewall-cmd --permanent --add-service=mysql
success
[root@system zabbix]# firewall-cmd --reload
success
[root@system zabbix]# firewall-cmd --list-all | grep mysql
  services: cockpit dhcpv6-client http https mysql ssh
```

- create database , user(note: handle permissions).

```
[root@system zabbix]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 28
Server version: 10.11.11-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement
.

MariaDB [(none)]> CREATE DATABASE studentdb;
Query OK, 1 row affected (0.003 sec)
```

To ensure the user can perform only allowed operations on the database.

% allows the user to connect from any host.

```
MariaDB [(none)]> CREATE USER 'studentuser'@'%' IDENTIFIED BY 'password';
Query OK, 0 rows affected (0.006 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON studentdb.* TO 'studentuser'@'%';
Query OK, 0 rows affected (0.005 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.005 sec)
```

```
MariaDB [(none)]> USE studentdb;
Database changed
MariaDB [studentdb]> CREATE TABLE students (
    ->     student_number VARCHAR(10) PRIMARY KEY,
    ->     first_name VARCHAR(50),
    ->     last_name VARCHAR(50),
    ->     program_enrolled VARCHAR(50),
    ->     graduation_year INT
    -> );
Query OK, 0 rows affected (0.079 sec)

MariaDB [studentdb]> INSERT INTO students (student_number, first_name, last_n
ame, program_enrolled, graduation_year) VALUES
    -> ('110-001', 'Allen', 'Brown', 'Mechanical', 2017),
    -> ('110-002', 'David', 'Brown', 'Mechanical', 2017),
    -> ('110-003', 'Mary', 'Green', 'Mechanical', 2018),
    -> ('110-004', 'Dennis', 'Green', 'Electrical', 2018),
    -> ('110-005', 'Joseph', 'Black', 'Electrical', 2018),
    -> ('110-006', 'Dennis', 'Black', 'Electrical', 2020),
    -> ('110-007', 'Ritchie', 'Salt', 'Computer Science', 2020),
    -> ('110-008', 'Robert', 'Salt', 'Computer Science', 2020),
    -> ('110-009', 'David', 'Suzuki', 'Computer Science', 2020),
    -> ('110-010', 'Mary', 'Chen', 'Computer Science', 2020);
Query OK, 10 rows affected (0.012 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
MariaDB [studentdb]> SELECT * FROM students;
+----------------+------------+-----------+------------------+-----------------+
| student_number | first_name | last_name | program_enrolled | graduation_year |
+----------------+------------+-----------+------------------+-----------------+
| 110-001        | Allen      | Brown     | Mechanical       |            2017 |
| 110-002        | David      | Brown     | Mechanical       |            2017 |
| 110-003        | Mary       | Green     | Mechanical       |            2018 |
| 110-004        | Dennis     | Green     | Electrical       |            2018 |
| 110-005        | Joseph     | Black     | Electrical       |            2018 |
| 110-006        | Dennis     | Black     | Electrical       |            2020 |
| 110-007        | Ritchie    | Salt      | Computer Science |            2020 |
| 110-008        | Robert     | Salt      | Computer Science |            2020 |
| 110-009        | David      | Suzuki    | Computer Science |            2020 |
| 110-010        | Mary       | Chen      | Computer Science |            2020 |
+----------------+------------+-----------+------------------+-----------------+
10 rows in set (0.001 sec)
```

- connect to the database created in step 3 using the new user (with password) Connecting with the new user verifies that the account works, has the correct permissions, and can access and modify the database as intended.

```
[root@system zabbix]# mysql -u studentuser -p -h 127.0.0.1 studentdb
Enter password:
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 31
Server version: 10.11.11-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement
.

MariaDB [studentdb]> SELECT * FROM students;
+----------------+------------+-----------+------------------+-----------------+
| student_number | first_name | last_name | program_enrolled | graduation_year |
+----------------+------------+-----------+------------------+-----------------+
| 110-001        | Allen      | Brown     | Mechanical       |            2017 |
| 110-002        | David      | Brown     | Mechanical       |            2017 |
| 110-003        | Mary       | Green     | Mechanical       |            2018 |
| 110-004        | Dennis     | Green     | Electrical       |            2018 |
| 110-005        | Joseph     | Black     | Electrical       |            2018 |
| 110-006        | Dennis     | Black     | Electrical       |            2020 |
| 110-007        | Ritchie    | Salt      | Computer Science |            2020 |
| 110-008        | Robert     | Salt      | Computer Science |            2020 |
| 110-009        | David      | Suzuki    | Computer Science |            2020 |
| 110-010        | Mary       | Chen      | Computer Science |            2020 |
+----------------+------------+-----------+------------------+-----------------+
10 rows in set (0.001 sec)
```

Implement a web page to show system statistics to the user

- Create directories

```
[root@system ~]# mkdir -p /var/www/html/sysstats/cpu
[root@system ~]# mkdir -p /var/www/html/sysstats/mem
[root@system ~]# mkdir -p /var/www/html/sysstats/disk
[root@system ~]# chown -R apache:apache /var/www/html/sysstats
[root@system ~]# chmod -R 755 /var/www/html/sysstats
```

- The collection script and Make the script executable

```
[root@system ~]# vim collect_stats.sh
```

-b → batch mode (output suitable for scripts)

-n1 → run **once** instead of interactive mode

```
#!/bin/bash
# Collect CPU, memory, disk stats and save to files with timestamp

STATS_DIR="/var/www/html/sysstats"
TIMESTAMP=$(date +"%Y%m%d%H%M")  # sortable timestamp

# CPU usage (user + system)
CPU=$(top -bn2 | grep "Cpu(s)" | tail -n1 | awk '{print $2+$4}')
echo $CPU > $STATS_DIR/cpu/cpu_$TIMESTAMP.txt

# Memory usage
MEM_USED=$(free -m | awk '/Mem:/ {print $3}')
MEM_FREE=$(free -m | awk '/Mem:/ {print $4}')
echo "$MEM_USED $MEM_FREE" > $STATS_DIR/memory/memory_$TIMESTAMP.txt

# Disk usage for root (/)
DISK_USED=$(df -h / | awk 'NR==2 {print $3}')
DISK_FREE=$(df -h / | awk 'NR==2 {print $4}')
echo "$DISK_USED $DISK_FREE" > $STATS_DIR/disk/disk_$TIMESTAMP.txt
```

```
[root@system ~]# chmod +x /root/collect_stats.sh
[root@system ~]# crontab -e
crontab: installing new crontab
Backup of root's previous crontab saved to /root/.cache/crontab/crontab.bak
```

```
# run collect_stats.sh at minute 0 of every hour
#0 * * * * /root/collect_stats.sh

# run every 5 minutes for testing
*/5 * * * * /root/collect_stats.sh
```

- The calculate script and Make the script executable

```
[root@system ~]# vim calculate_avg.sh
[root@system ~]# chmod +x /root/calculate_avg.sh
```

```
#!/bin/bash

# Directories
CPU_DIR="/var/www/html/sysstats/cpu"
MEM_DIR="/var/www/html/sysstats/memory"
DISK_DIR="/var/www/html/sysstats/disk"

# CPU average
awk '{sum+=$1; count++} END {if(count>0) print sum/count}' $CPU_DIR/cpu_*.txt
 > $CPU_DIR/avg_cpu.txt

# Memory average (used and free)
awk '{sum_used+=$1; sum_free+=$2; count++} END {if(count>0) print "Used Avg:"
, sum_used/count, "Free Avg:", sum_free/count}' $MEM_DIR/memory_*.txt > $MEM_
DIR/avg_memory.txt

# Disk average (used and free)
awk '{sum_used+=$1; sum_free+=$2; count++} END {if(count>0) print "Used Avg:"
, sum_used/count, "Free Avg:", sum_free/count}' $DISK_DIR/disk_*.txt > $DISK_
DIR/avg_disk.txt
```

```
# run collect_stats.sh at minute 0 of every hour
#0 * * * * /root/collect_stats.sh

# run every 5 minutes for testing
* * * * * /root/collect_stats.sh

# run ccalculate_avg.sh 5 minutes
5 * * * * /root/calculate_avg.sh

# run every 5 minutes for testing
* * * * * /root/calculate_avg.sh
```

- Install Apache

  Apache is a widely-used web server that serves static and dynamic web content to clients over HTTP/HTTPS. In this task, Apache is required to display system statistics on a web page for users to access.

```
[root@system ~]# yum install httpd -y
Repository AppStream is listed more than once in the configuration
Last metadata expiration check: 1:55:46 ago on Tue 02 Dec 2025 01:24:25 PM EE
T.
Package httpd-2.4.63-1.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@system ~]# systemctl start httpd
[root@system ~]# systemctl enable httpd
```

  curl http://localhost/sysstats/

  I access it via `http://localhost:8080/sysstats/index.html`

  because I set up port forwarding in VirtualBox from host port 8080 to the guest's Apache port 80.

  This is when I created separate files for HTML without including them with shell script.

System Statistics

Last updated: 2025-12-02T11:05:02Z (UTC)

CPU Usage — avg: 8.80% (samples: 9)

Memory Usage — avg used: 2903.56 MB, avg free: 341.00 MB (samples: 9)

Disk Usage — avg used: 11755.85 MB (samples: 9)

- Prepare the HTML structure

```
#!/bin/bash
STATS_DIR="/var/www/html/sysstats"
TIMESTAMP=$(date +"%Y%m%d%H%M")

# CPU usage
CPU=$(grep 'cpu ' /proc/stat | awk '{usage=($2+$4)*100/($2+$4+$5); printf "%.
1f", usage}')
echo $CPU > $STATS_DIR/cpu/cpu_$TIMESTAMP.txt

# Memory usage
MEM_USED=$(free -m | awk '/Mem:/ {print $3}')
MEM_FREE=$(free -m | awk '/Mem:/ {print $4}')
echo "$MEM_USED $MEM_FREE" > $STATS_DIR/memory/memory_$TIMESTAMP.txt

# Disk usage
DISK_USED=$(df -m / | awk 'NR==2 {print $3}')
DISK_FREE=$(df -m / | awk 'NR==2 {print $4}')
echo "$DISK_USED $DISK_FREE" > $STATS_DIR/disk/disk_$TIMESTAMP.txt

# --- Optional: regenerate HTML files ---
/root/calculate_avg.sh
```

```
#!/bin/bash

STATS_DIR="/var/www/html/sysstats"
TIMESTAMP=$(date +"%Y-%m-%d %H:%M:%S")

# ---------------- CPU ----------------
CPU_FILES=($STATS_DIR/cpu/cpu_*.txt)
CPU_SUM=0
CPU_COUNT=0
for f in "${CPU_FILES[@]}"; do
    val=$(cat "$f")
    # Make sure it's a number
    if [[ $val =~ ^[0-9.]+$ ]]; then
        CPU_SUM=$(echo "$CPU_SUM + $val" | bc)
        CPU_COUNT=$((CPU_COUNT + 1))
    fi
done
if [ $CPU_COUNT -gt 0 ]; then
    CPU_AVG=$(echo "scale=2; $CPU_SUM / $CPU_COUNT" | bc)
else
    CPU_AVG=0
fi
echo "$CPU_AVG" > $STATS_DIR/cpu/avg_cpu.txt

# ---------------- Memory ----------------
MEM_FILES=($STATS_DIR/memory/memory_*.txt)
MEM_USED_SUM=0
MEM_FREE_SUM=0
MEM_COUNT=0
for f in "${MEM_FILES[@]}"; do
    read used free < "$f"
    # Make sure both are numbers
    if [[ $used =~ ^[0-9]+$ && $free =~ ^[0-9]+$ ]]; then
        MEM_USED_SUM=$((MEM_USED_SUM + used))
                                                        1,1           Top
```

```
        MEM_FREE_SUM=$((MEM_FREE_SUM + free))
        MEM_COUNT=$((MEM_COUNT+1))
    fi
done
if [ $MEM_COUNT -gt 0 ]; then
    MEM_USED_AVG=$(echo "scale=2; $MEM_USED_SUM / $MEM_COUNT" | bc)
    MEM_FREE_AVG=$(echo "scale=2; $MEM_FREE_SUM / $MEM_COUNT" | bc)
else
    MEM_USED_AVG=0
    MEM_FREE_AVG=0
fi
echo "$MEM_USED_AVG $MEM_FREE_AVG" > $STATS_DIR/memory/avg_memory.txt

# ---------------- Disk ----------------
DISK_FILES=($STATS_DIR/disk/disk_*.txt)
DISK_USED_SUM=0
DISK_FREE_SUM=0
DISK_COUNT=0
for f in "${DISK_FILES[@]}"; do
    read used free < "$f"
    if [[ $used =~ ^[0-9]+$ && $free =~ ^[0-9]+$ ]]; then
        DISK_USED_SUM=$((DISK_USED_SUM + used))
        DISK_FREE_SUM=$((DISK_FREE_SUM + free))
        DISK_COUNT=$((DISK_COUNT+1))
    fi
done
if [ $DISK_COUNT -gt 0 ]; then
    DISK_USED_AVG=$(echo "scale=2; $DISK_USED_SUM / $DISK_COUNT" | bc)
    DISK_FREE_AVG=$(echo "scale=2; $DISK_FREE_SUM / $DISK_COUNT" | bc)
else
    DISK_USED_AVG=0
    DISK_FREE_AVG=0
fi
echo "$DISK_USED_AVG $DISK_FREE_AVG" > $STATS_DIR/disk/avg_disk.txt
                                                          40,1          27%
```

```
# ---------------- Generate HTML pages ----------------

# Main index.html
cat > $STATS_DIR/index.html <<EOL
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<title>System Statistics</title>
<style>
body { font-family: Arial, sans-serif; max-width:900px; margin:2rem auto; pad
ding:0 1rem; }
h1 { color:#222; }
ul { list-style: none; padding:0;}
li { margin: 0.5rem 0; }
a { text-decoration: none; color: #0066cc; }
.meta { color: #666; font-size: 0.9rem;}
</style>
</head>
<body>
<h1>System Statistics</h1>
<p class="meta">Last updated: $TIMESTAMP</p>
<ul>
<li><a href="cpu.html">CPU Usage</a> — avg: $CPU_AVG% (samples: $CPU_COUNT)</
li>
<li><a href="mem.html">Memory Usage</a> — avg used: $MEM_USED_AVG MB, avg fre
e: $MEM_FREE_AVG MB (samples: $MEM_COUNT)</li>
<li><a href="disk.html">Disk Usage</a> — avg used: $DISK_USED_AVG MB, avg fre
e: $DISK_FREE_AVG MB (samples: $DISK_COUNT)</li>
</ul>
</body>
</html>
EOL
```

```
# CPU page
cat > $STATS_DIR/cpu.html <<EOL
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<title>CPU Usage</title>
</head>
<body>
<h1>CPU Usage</h1>
<p>Average: $CPU_AVG%</p>
<ul>
EOL
for f in "${CPU_FILES[@]}"; do
    ts=$(basename "$f" | sed 's/cpu_//; s/.txt//')
    val=$(cat "$f")
    echo "<li>$ts — $val%</li>" >> $STATS_DIR/cpu.html
done
echo "</ul></body></html>" >> $STATS_DIR/cpu.html

# Memory page
cat > $STATS_DIR/mem.html <<EOL
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<title>Memory Usage</title>
</head>
<body>
<h1>Memory Usage</h1>
<p>Average Used: $MEM_USED_AVG MB, Average Free: $MEM_FREE_AVG MB</p>
<ul>
```

```
for f in "${MEM_FILES[@]}"; do
    ts=$(basename "$f" | sed 's/memory_//; s/.txt//')
    read used free < "$f"
    echo "<li>$ts — Used: $used MB, Free: $free MB</li>" >> $STATS_DIR/mem.ht
ml
done
echo "</ul></body></html>" >> $STATS_DIR/mem.html

# Disk page
cat > $STATS_DIR/disk.html <<EOL
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<title>Disk Usage</title>
</head>
<body>
<h1>Disk Usage</h1>
<p>Average Used: $DISK_USED_AVG MB, Average Free: $DISK_FREE_AVG MB</p>
<ul>
EOL
for f in "${DISK_FILES[@]}"; do
    ts=$(basename "$f" | sed 's/disk_//; s/.txt//')
    read used free < "$f"
    echo "<li>$ts — Used: $used MB, Free: $free MB</li>" >> $STATS_DIR/disk.h
tml
done
echo "</ul></body></html>" >> $STATS_DIR/disk.html
```

System Statistics

Last updated: 2025-12-02 16:20:02

CPU Usage — avg: 8.73% (samples: 153)

Memory Usage — avg used: 1432.00 MB, avg free: 1771.63 MB (samples: 41)

Disk Usage — avg used: 11807.00 MB, avg free: 13730.00 MB (samples: 5)

CPU Usage

Average: 8.73%

- 202511280150 — 15.1%
- 202511280151 — 20%
- 202511280152 — 0%
- 202511280153 — 0%
- 202511280154 — 14.8%
- 202511280155 — 4.5%
- 202511280156 — 0%
- 202511280157 — 0%
- 202511280158 — 0%
- 202511280159 — 0%
- 202511280200 — 8.6%

Disk Usage

Average Used: 11806.00 MB, Average Free: 13731.00 MB

- 202512021719 — Used: 11806 MB, Free: 13731 MB
- 202512021720 — Used: 11806 MB, Free: 13731 MB
- 202512021725 — Used: 11806 MB, Free: 13731 MB